

SCR2043 OPERATING SYSTEMS

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Section : 03

Marks

This lab assessment is designed to test your understanding and skills on some basic concepts and tools related to process monitoring and management in operating system. Please follow the instructions carefully and submit your answers in this word document and rename the file as **os-lab-assessment02-studentname-matricno.docx**.

Essential Steps Before Starting Lab Assessment 2:

1. Download necessary source codes:

Use the `wget` command to retrieve the following source code files to your Linux (or WSL or MacOS) environment:

```
wget -O mainprocess.c https://rebrand.ly/mainprocess_c
wget -O subprocess1.c https://rebrand.ly/subprocess1_c
wget -O subprocess2.c https://rebrand.ly/subprocess2_c
```

2. Compile the source files:

Use the `gcc` compiler to create executable files from the source code.

```
gcc mainprocess.c -o mainprocess
gcc subprocess1.c -o subprocess1
gcc subprocess2.c -o subprocess2
```

3. Execute the dummy processes:

Run all the dummy processes

```
./mainprocess &
```

Press **enter** two times.

4. The dummy processes are running for 2 hours. If you took longer than 2 hours on questions 1-9, please restart the main process with `./mainprocess &`.

Lab Assessment 2: Linux Process Monitoring and Management

Instructions:

1. Carefully execute each command as instructed in the questions.
2. Write down the exact command used for each task.
3. Capture a screenshot of the command's output.

Question 1

Use the `ps` command with the appropriate option to display a complete list of all running processes within the Linux operating system.

Command	
<code>ps -e</code>	
Output	
<pre> 83 ? 00:00:00 kworker/R-crypt 93 ? 00:00:00 kworker/R-charg 132 ? 00:00:00 kworker/u4:1-ext4-rsv-conversion 147 ? 00:00:00 kworker/R-mpt_p 148 ? 00:00:00 kworker/R-mpt/0 156 ? 00:00:00 scsi_eh_2 157 ? 00:00:00 kworker/R-scsi_ 191 ? 00:00:00 kworker/R-raid5 214 ? 00:00:00 kworker/0:2H-kblockd 233 ? 00:00:00 jbd2/sda1-8 234 ? 00:00:00 kworker/R-ext4- 308 ? 00:00:00 systemd-journal 328 ? 00:00:00 kworker/R-kmpat 329 ? 00:00:00 kworker/R-kmpat 353 ? 00:00:00 multipathd 377 ? 00:00:00 systemd-udevd 378 ? 00:00:00 psimon 449 ? 00:00:00 irq/18-vmwgfx 452 ? 00:00:00 kworker/R-ttm 478 ? 00:00:00 jbd2/sda16-8 479 ? 00:00:00 kworker/R-ext4- 516 ? 00:00:00 systemd-network 521 ? 00:00:00 systemd-resolve 525 ? 00:00:00 systemd-timesyn 640 ? 00:00:00 dbus-daemon 649 ? 00:00:00 polkitd 657 ? 00:00:00 systemd-logind 661 ? 00:00:00 udisksd 679 ? 00:00:00 rsyslogd 706 ? 00:00:00 cron 716 ? 00:00:01 unattended-upgr 728 ? 00:00:00 ModemManager 746 tty1 00:00:00 login 843 ? 00:00:00 kworker/1:2H-kblockd 962 ? 00:00:00 psimon 964 ? 00:00:00 systemd</pre>	

965	?	00:00:00	(sd-pam)
976	tty1	00:00:00	bash
1017	?	00:00:00	kworker/R-tls-s
1046	tty1	00:00:00	mainprocess
1047	tty1	00:00:00	mainprocess
1048	tty1	00:00:00	subprocess1
1049	tty1	00:00:00	subprocess1
1050	tty1	00:00:00	mainprocess
1051	tty1	00:00:00	subprocess2
1052	tty1	00:00:00	subprocess2
1053	tty1	00:00:00	subprocess2
1054	?	00:00:00	kworker/u6:3-events_unbound
1055	tty1	00:00:00	ps

Question 2

Employ the `ps` command with necessary options to unveil comprehensive details about each running process.

Command					
<code>ps -ef</code>					
Output					
root	449	2	0 14:57 ?	00:00:00	[irq/16-vmmgfx]
root	452	2	0 14:57 ?	00:00:00	[kworker/R-ttm]
root	476	2	0 14:57 ?	00:00:00	[jbd2/sda16-8]
root	479	2	0 14:57 ?	00:00:00	[kworker/R-ext4-]
systemd+	516	1	0 14:57 ?	00:00:00	/usr/lib/systemd/systemd-networkd
systemd+	521	1	0 14:57 ?	00:00:00	/usr/lib/systemd/systemd-resolved
systemd+	525	1	0 14:57 ?	00:00:00	/usr/lib/systemd/systemd-timesyncd
message+	640	1	0 14:57 ?	00:00:00	dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --syslog-only
polkitd	649	1	0 14:57 ?	00:00:00	/usr/lib/polkit-1/polkitd --no-debug
root	657	1	0 14:57 ?	00:00:00	/usr/lib/systemd/systemd-logind
root	661	1	0 14:57 ?	00:00:00	/usr/libexec/udisks2/udisksd
syslog	679	1	0 14:57 ?	00:00:00	/usr/sbin/rsyslogd -n -iNONE
root	706	1	0 14:57 ?	00:00:00	/usr/sbin/cron -f -P
root	716	1	0 14:57 ?	00:00:01	/usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal
root	728	1	0 14:57 ?	00:00:00	/usr/sbin/ModemManager
root	746	1	0 14:57 tty1	00:00:00	/bin/login -p --
root	843	2	0 14:57 ?	00:00:00	[kworker/1:2H-kblockd]
root	962	2	0 14:58 ?	00:00:00	[psimon]
adriana	964	1	0 14:58 ?	00:00:00	/usr/lib/systemd/systemd --user
adriana	965	964	0 14:58 ?	00:00:00	(sd-pam)
adriana	976	746	0 14:58 tty1	00:00:00	-bash
root	1017	2	0 14:58 ?	00:00:00	[kworker/R-tls-s]
adriana	1046	976	0 15:05 tty1	00:00:00	./mainprocess
adriana	1047	1046	0 15:05 tty1	00:00:00	./mainprocess
adriana	1048	1047	0 15:05 tty1	00:00:00	./subprocess1
adriana	1049	1047	0 15:05 tty1	00:00:00	./subprocess1
adriana	1050	1046	0 15:05 tty1	00:00:00	./mainprocess
adriana	1051	1050	0 15:05 tty1	00:00:00	./subprocess2
adriana	1052	1050	0 15:05 tty1	00:00:00	./subprocess2
adriana	1053	1050	0 15:05 tty1	00:00:00	./subprocess2
root	1054	2	0 15:05 ?	00:00:00	[kworker/u6:3-events_unbound]
root	1060	2	1 15:09 ?	00:00:07	[kworker/0:0-events]
root	1063	2	0 15:10 ?	00:00:00	[kworker/1:2-events]
root	1064	2	0 15:11 ?	00:00:00	[kworker/u6:2-events_power_efficient]
root	1073	1	1 15:12 ?	00:00:06	/usr/libexec/fwupd/fwupd
root	1106	2	0 15:12 ?	00:00:00	[kworker/1:1H]
root	1108	2	0 15:13 ?	00:00:00	[kworker/u5:2-events_power_efficient]
root	1109	2	0 15:13 ?	00:00:00	[kworker/0:1H]
root	1114	2	1 15:16 ?	00:00:03	[kworker/1:1-events]
root	1123	2	0 15:20 ?	00:00:00	[kworker/0:1-events]
adriana	1129	976	0 15:20 tty1	00:00:00	./mainprocess
adriana	1130	1129	0 15:20 tty1	00:00:00	./mainprocess
adriana	1131	1130	0 15:20 tty1	00:00:00	./subprocess1
adriana	1132	1130	0 15:20 tty1	00:00:00	./subprocess1
adriana	1133	1129	0 15:20 tty1	00:00:00	./mainprocess
adriana	1134	1133	0 15:20 tty1	00:00:00	./subprocess2
adriana	1135	1133	0 15:20 tty1	00:00:00	./subprocess2
adriana	1136	1133	0 15:20 tty1	00:00:00	./subprocess2
adriana	1137	976	99 15:20 tty1	00:00:00	ps -ef

Question 3

Use the `ps` command with some tools to only list processes named "subprocess" and show some info about them.

Command
<code>ps -e grep subprocess</code>
Output
<pre>adriana@secr2043:~\$ ps -e grep subprocess 1048 tty1 00:00:00 subprocess1 1049 tty1 00:00:00 subprocess1 1051 tty1 00:00:00 subprocess2 1052 tty1 00:00:00 subprocess2 1053 tty1 00:00:00 subprocess2 1131 tty1 00:00:00 subprocess1 1132 tty1 00:00:00 subprocess1 1134 tty1 00:00:00 subprocess2 1135 tty1 00:00:00 subprocess2 1136 tty1 00:00:00 subprocess2</pre>

Question 4

Execute the `ps` command, specifying options that reveal only the following columns:

- Process ID (`pid`)
- Owner of the process (`user`)
- CPU percentage (`pcpu`)
- Memory percentage (`pmem`)
- Command (`cmd`)

Command
<code>ps -eo pid,user,pcpu,pmem,cmd</code>
Output

```

521 systemd+ 0.0 1.2 /usr/lib/systemd/systemd-resolved
525 systemd+ 0.0 0.7 /usr/lib/systemd/systemd-timesyncd
640 message+ 0.0 0.5 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --syslog-only
649 polkitd 0.0 0.8 /usr/lib/polkit-1/polkitd --no-debug
657 root 0.0 0.8 /usr/lib/systemd/systemd-logind
661 root 0.0 1.3 /usr/libexec/udisks2/udisksd
679 syslog 0.0 0.6 /usr/sbin/rsyslogd -n -iNONE
706 root 0.0 0.2 /usr/sbin/cron -f -P
716 root 0.0 2.3 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal
728 root 0.0 1.3 /usr/sbin/ModemManager
746 root 0.0 0.4 /bin/login -p --
962 root 0.0 0.0 [psimon]
964 adriana 0.0 1.1 /usr/lib/systemd/systemd --user
965 adriana 0.0 0.3 (sd-pam)
976 adriana 0.0 0.5 -bash
1017 root 0.0 0.0 [kworker/R-tls-s]
1046 adriana 0.0 0.1 ./mainprocess
1047 adriana 0.0 0.0 ./mainprocess
1048 adriana 0.0 0.1 ./subprocess1
1049 adriana 0.0 0.1 ./subprocess1
1050 adriana 0.0 0.0 ./mainprocess
1051 adriana 0.0 0.1 ./subprocess2
1052 adriana 0.0 0.1 ./subprocess2
1053 adriana 0.0 0.1 ./subprocess2
1073 root 0.4 4.1 /usr/libexec/fwupd/fwupd
1106 root 0.0 0.0 [kworker/1:1H]
1109 root 0.0 0.0 [kworker/0:1H]
1114 root 1.2 0.0 [kworker/1:1-events]
1123 root 1.0 0.0 [kworker/0:1-events]
1129 adriana 0.0 0.1 ./mainprocess
1130 adriana 0.0 0.0 ./mainprocess
1131 adriana 0.0 0.1 ./subprocess1
1132 adriana 0.0 0.1 ./subprocess1
1133 adriana 0.0 0.0 ./mainprocess
1134 adriana 0.0 0.1 ./subprocess2
1135 adriana 0.0 0.1 ./subprocess2
1136 adriana 0.0 0.1 ./subprocess2
1138 root 0.0 0.0 [kworker/u5:0-events_unbound]
1139 root 0.0 0.0 [kworker/u6:0-events_power_efficient]
1145 root 0.0 0.0 [kworker/u5:2-events_unbound]
1148 root 0.1 0.0 [kworker/1:2-events]
1149 root 0.7 0.0 [kworker/0:3-mpt_poll_0]
1150 root 0.0 0.0 [kworker/u6:2-events_unbound]
1154 root 0.0 0.0 [kworker/1:0H-kblockd]
1155 root 1.6 0.0 [kworker/1:0-events]
1159 root 0.0 0.0 [kworker/0:0H-kblockd]
1160 root 0.5 0.0 [kworker/0:0-ata_sff]
1164 root 0.0 0.0 [kworker/u5:3-flush-8:0]
1166 adriana 100 0.4 ps -eo pid,user,pcpu,pmem,cmd

```

Question 5

Building on the ps command used in Question 4, can you add an option to sort the listed processes by their memory usage (pmem)?

Command
<code>ps -eo pid,user,pcpu,pmem,cmd --sort=-pmem</code>
Output

```

52 root      0.0  0.0 [watchdogd]
54 root      0.0  0.0 [kswapd0]
55 root      0.0  0.0 [ecryptfs-kthread]
56 root      0.0  0.0 [kworker/R-kthrd]
57 root      0.0  0.0 [kworker/R-acpi_]
58 root      0.0  0.0 [scsi_eh_0]
59 root      0.0  0.0 [kworker/R-scsi_]
60 root      0.0  0.0 [scsi_eh_1]
61 root      0.0  0.0 [kworker/R-scsi_]
65 root      0.0  0.0 [kworker/R-mld]
66 root      0.0  0.0 [kworker/R-ipv6_]
74 root      0.0  0.0 [kworker/R-kstrp]
76 root      0.0  0.0 [kworker/u7:0]
77 root      0.0  0.0 [kworker/u8:0]
78 root      0.0  0.0 [kworker/u9:0]
83 root      0.0  0.0 [kworker/R-crypt]
93 root      0.0  0.0 [kworker/R-charg]
132 root     0.0  0.0 [kworker/u4:1-ext4-rsv-conversion]
147 root     0.0  0.0 [kworker/R-mpt_p]
148 root     0.0  0.0 [kworker/R-mpt/0]
156 root     0.0  0.0 [scsi_eh_2]
157 root     0.0  0.0 [kworker/R-scsi_]
191 root     0.0  0.0 [kworker/R-raid5]
233 root     0.0  0.0 [jbd2/sda1-8]
234 root     0.0  0.0 [kworker/R-ext4-]
328 root     0.0  0.0 [kworker/R-kmpat]
329 root     0.0  0.0 [kworker/R-kmpat]
378 root     0.0  0.0 [psimon]
449 root     0.0  0.0 [irq/18-vmwgfx]
452 root     0.0  0.0 [kworker/R-ttm]
478 root     0.0  0.0 [jbd2/sda16-8]
479 root     0.0  0.0 [kworker/R-ext4-]
962 root     0.0  0.0 [psimon]
1017 root    0.0  0.0 [kworker/R-tls-s]
1106 root    0.0  0.0 [kworker/1:1H]
1109 root    0.0  0.0 [kworker/0:1H]
1114 root    1.2  0.0 [kworker/1:1-events]
1123 root    1.0  0.0 [kworker/0:1-events]
1138 root    0.0  0.0 [kworker/u5:0-events_unbound]
1139 root    0.0  0.0 [kworker/u6:0-events_unbound]
1145 root    0.0  0.0 [kworker/u5:2-events_unbound]
1150 root    0.0  0.0 [kworker/u6:2-events_power_efficient]
1154 root    0.0  0.0 [kworker/1:0H-kblockd]
1155 root    0.5  0.0 [kworker/1:0-events]
1159 root    0.0  0.0 [kworker/0:0H-kblockd]
1164 root    0.0  0.0 [kworker/u5:3-flush-8:0]
1169 root    0.0  0.0 [kworker/u6:3-events_power_efficient]
1171 root    0.4  0.0 [kworker/0:2-events]
1172 root    0.0  0.0 [kworker/1:2]

```

Question 6

Construct a command using `ps`, suitable options, and any additional tools to visualize the hierarchical structure (tree-like) of the following processes:

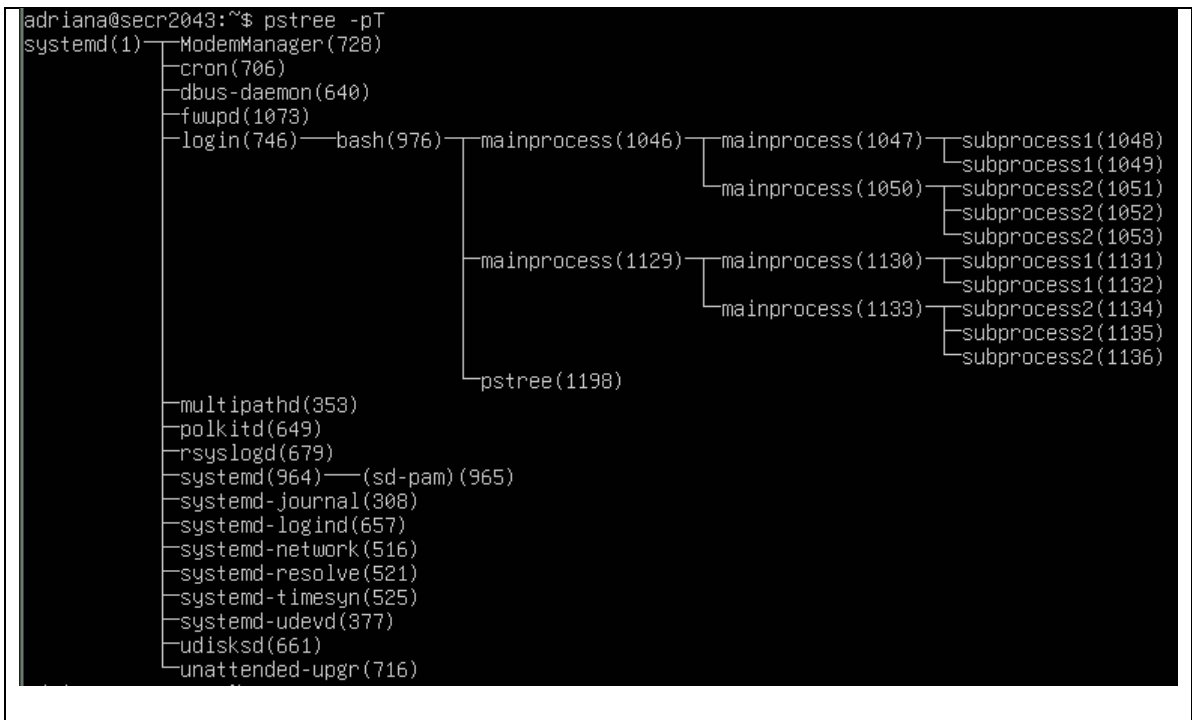
- "mainprocess"
- "subprocess1"
- "subprocess2"

Command
<code>ps -forest -C mainprocess -C subprocess1 -C subprocess2</code>
Output
<pre> adriana@secr2043:~\$ ps --forest -C mainprocess -C subprocess1 -C subprocess2 PID TTY TIME CMD 1382 tty1 00:00:00 mainprocess 1383 tty1 00:00:00 _ mainprocess 1385 tty1 00:00:00 _ subprocess1 1386 tty1 00:00:00 _ subprocess1 1384 tty1 00:00:00 _ mainprocess 1387 tty1 00:00:00 _ subprocess2 1388 tty1 00:00:00 _ subprocess2 1389 tty1 00:00:00 _ subprocess2 </pre>

Question 7

Use `pstree` command with option that show the number of threads to each process.

Command
<code>pstree -pT</code>
Output



Question 8

Use renice command to change priority level of one of process “subprocess1”.

Command
<pre>ps -e grep subprocess1 sudo renice -n 10 -p 1048</pre>
Output
<pre>adriana@secr2043:~\$ ps -e grep subprocess1 1048 tty1 00:00:00 subprocess1 1049 tty1 00:00:00 subprocess1 1131 tty1 00:00:00 subprocess1 1132 tty1 00:00:00 subprocess1 adriana@secr2043:~\$ sudo renice -n 10 -p 1048 [sudo] password for adriana: 1048 (process ID) old priority 0, new priority 10</pre>

Question 9

Terminate all running processes with the name “mainprocess”.

Command
<pre>pkill mainprocess</pre>

Output
<pre>adriana@secr2043:~\$ pkill mainprocess Main process (ID: 1133) received signal: 15. Terminating... Main process (ID: 1130) received signal: 15. Terminating... Main process (ID: 1129) received signal: 15. Terminating... [1]- Done ./mainprocess [2]+ Done ./mainprocess</pre>

Question 10

Write a short C or Python code (choose only one language) demonstrating multiprocessing with `fork()` and `wait()`. Compile and/or run the code. Show the output.

Source Code:

```
import os

pid = os.fork()

if pid == 0:
    print("Child process: PID =", os.getpid())
else:
    os.wait() # Wait for the child to finish
    print("Parent process: PID =", os.getpid())
```

Output:

```
adriana@secr2043:~$ python3 fork_example.py
Child process: PID = 1341
Parent process: PID = 1340
```